How-To Library

Informatica

Importing Metadata From a Teradata Source in Test Data Management

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Abstract

You can perform TDM operations on Teradata sources in Test Data Management (TDM).

This article describes the steps to import metadata from a Teradata database and to create a Teradata connection in Test Data Manager. This article assumes that you have knowledge of TDM and are familiar with TDM operations.

Supported Versions

- Test Data Management 10.2.0
- Test Data Management 10.1.0
- Test Data Management 9.7.1 HotFix 1
- Test Data Management 9.7.1
- Test Data Management 9.7.0 with EBF 16748 or later

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Overview

To perform TDM operations on source data, you must import the source metadata into the TDM repository.

You can perform TDM operations on data from a Teradata database.

You can create the following Teradata connections in Test Data Manager:

- Teradata
- Teradata FastExport. Use as a source connection only.
- Teradata FastLoad. Use as a target connection only.
- Teradata Multiload. Use as a target connection only.
- Teradata Parallel Transporter

Create connections to perform TDM operations on source data. When you create a project, add one or more sources to the project. You can then create constraints to create relationships between the sources and apply filter criteria to perform data subset operations. You can apply data masking or generation rules to perform data masking or data generation operations.

Rules and Guidelines

When you use a Teradata source in a TDM operation, consider the following rules and guidelines:

- You cannot import metadata from a Teradata data source directly into Test Data Manager. You must import the metadata from PowerCenter®.
- To use a Teradata or a Teradata Parallel Transporter connection as a staging connection, the `tdgssconfig.jar` and `terajdbc4.jar` Teradata JDBC JAR files must be present in the following location: `<Informatica installation directory>\TDM\utilities\mapgen\thirdpartylib`.

  Create a `thirdpartylib` folder and copy the JARs files before you configure the connection as a staging connection.
- You must perform the following tasks before you can test a Teradata or Teradata Parallel Transporter connection during connection creation:
  On versions 9.7.1 Hot Fix 1, 10.1.0 and 10.2.0

  Add the `tdgssconfig.jar` and `terajdbc4.jar` Teradata JDBC JAR files to the `<Informatica installation directory>\TDM\lib\thirdparty` location and restart the Test Data Manager Service.

  On versions 9.7.0 with EBF 16748 or later and 9.7.1

  1. Add the `tdgssconfig.jar` and `terajdbc4.jar` Teradata JDBC JAR files to the `<Informatica installation directory>\TDM\lib` location.
  2. Add the following entries to the `<Informatica installation directory>\TDM\configuration\java7-server.profile` file at the end of the following section:
     org.osgi.framework.system.packages = \

     ```
     com.ncr.teradata,\
     com.teradata.jdbc,\
     com.teradata.jdbc.client,\
     com.teradata.jdbc.encode,\
     com.teradata.jdbc.jdbc,\
     com.teradata.jdbc.jdbc.console,\
     com.teradata.jdbc.jdbc.fastexport,\
     com.teradata.jdbc.jdbc.fastload,\
     com.teradata.jdbc.jdbc.monitor,\
     com.teradata.jdbc.jdbc.raw,\
     com.teradata.jdbc.jdbc_3.dbmetadata,\
     com.teradata.jdbc.jdbc_4,\
     com.teradata.jdbc.jdbc_4.ifsupport,\
     com.teradata.jdbc.jdbc_4.io,\
     com.teradata.jdbc.jdbc_4.logging,\
     com.teradata.jdbc.jdbc_4.parcel,\
     com.teradata.jdbc.jdbc_4.statemachine,\
     com.teradata.jdbc.jdbc_4.util,\
     com.teradata.jdbc.jdk14,\
     com.teradata.jdbc.jdk6,\
     com.teradata.jdbc.resource,\
     com.teradata.tdgss.jalgapi,\
     com.teradata.tdgss.jgssp2gss,\
     com.teradata.tdgss.jgssp2ldap,\
     com.teradata.tdgss.jgssp2td1,\
     com.teradata.tdgss.jgssp2td2,\
     com.teradata.tdgss.jgsspi,\
     com.teradata.tdgss.jtdgss
     ```
  3. Restart the Test Data Manager Service.

  You cannot test a Teradata or Teradata Parallel Transporter connection when you create the connection in Test Data Manager before you copy and update the files.

Example Scenario

You work with a software product testing team that creates and manages its test data in TDM.

You need to create test data from a Teradata data source. To use data from a Teradata database, you import the metadata into Test Data Manager. You can then run a TDM operation to create the test data.

To import metadata from a Teradata data source into Test Data Manager, perform the following tasks:

1. Configure a 32-bit data source name (DSN) with 32-bit Teradata ODBC drivers on the PowerCenter Client machine.
2. Configure a 64-bit DSN with 64-bit Teradata ODBC drivers on the machine on which you install Informatica services.
3. Import the source metadata into PowerCenter using the 32-bit DSN that you created in step 1.
4. Create a project in Test Data Manager and import the source metadata from the PowerCenter repository.
5. Create a Teradata connection in Test Data Manager. You use the connection in a plan to perform TDM operations.

Configure a 32-bit Teradata DSN

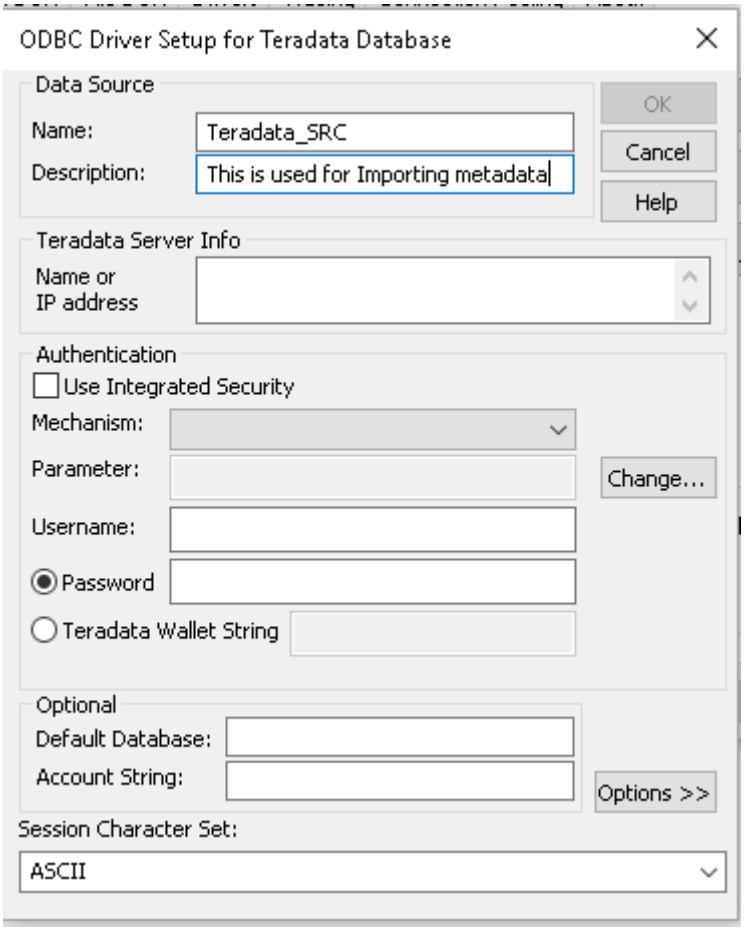
PowerCenter Client requires a 32-bit Teradata ODBC DSN to import metadata from a Teradata database. To import metadata into PowerCenter, configure a 32-bit Teradata ODBC data source that connects to a Teradata server on the PowerCenter Client machine.

Install the Teradata 32-bit client before you configure the DSN.

1. On the PowerCenter Client machine, open ODBC Data Source Administrator (32-bit).
2. Click **System DSN** > **Add**.

   The **Create New Data Source** window opens.
3. Select the Teradata driver and click **Finish**.
4. In the **ODBC Driver Setup for Teradata Database** window, enter a data source name and description. Enter the server, port, and other database details.

The following image shows the **ODBC Driver Setup for Teradata Database** window:



5. Test the connection to validate the information.

   The data source is created and appears in the list of system data sources.

6. Click **OK**.

Configure a 64-bit Teradata ODBC DSN

The PowerCenter Integration Service requires a 64-bit data source to perform data movement. Configure a 64-bit Teradata ODBC data source for each Teradata database that you want to access to perform TDM operations. You can create a 64-bit Teradata ODBC data source on Windows and Linux machines. This task describes the steps to configure the data source name on a Linux machine.

Install the Linux Teradata 64-bit client on the machine on which you create the PowerCenter Integration Service.

1. Open the `<Informatica install directory>\ODBC7.1\odbc.ini` file.
2. Add an entry for each data source that you create in the [ODBC Data Sources] section. For example, add the following in the [ODBC Data Sources] section:

   ```
   Teradata=tdata.so
   ```

3. Configure the data source properties to connect to the Teradata database. You use the connections to perform TDM operations. Add an entry with the following properties for each Teradata connection that you create:

```
[Teradata]
Driver=/opt/teradata/client/15.10/lib64/tdata.so
Description=
DBCName=
LastUser=
Username=
Password=
Database=
DateTimeFormat=
DefaultDatabase=
```

4. Save and close the file.
5. Set the following environment variables under .bash_profile.
   - `export TERADATA_HOME=/opt/teradata/client/15.10`
   - `export MANPATH=/opt/teradata/client/15.10/odbc_64/help/man:/opt/teradata/client/15.10/odbc_64/help/man`
   - `export ODBCINI=$ODBCHOME/odbc.ini`
   - `export TWB_ROOT=/opt/teradata/client/15.10/tbuild`
   - `export NLSPATH=/opt/teradata/client/15.10/msg/%N:/opt/teradata/client/15.10/odbc_64/msg/%N:/usr/lib/nls/msg/%L/%N:/usr/lib/nls/msg/%L/%N.cat`
   - `export TD_ICU_DATA=/opt/teradata/client/15.10/tdicu`
   - `export TD_ODBC_HOME=/opt/teradata/client/ODBC_64`
6. Add the following entry to LD_LIBRARY_PATH. Maintain the order of the entries.

   `/opt/teradata/client/15.10/lib64:$ODBCHOME/lib:/opt/ibm/db2/V10.5/lib64:$ORACLE_HOME/lib:$INFA_HOME/server/bin:/usr/lib64:/usr/lib64:/opt/teradata/client/15.10/odbc_64/msg:/opt/teradata/client/15.10/tbuild/lib64:/opt/teradata/teragss/linux-x8664/15.10.00.00/lib:/opt/teradata/client/15.10/lib64:/opt/teradata/client/15.10/lib64:/usr/lib`

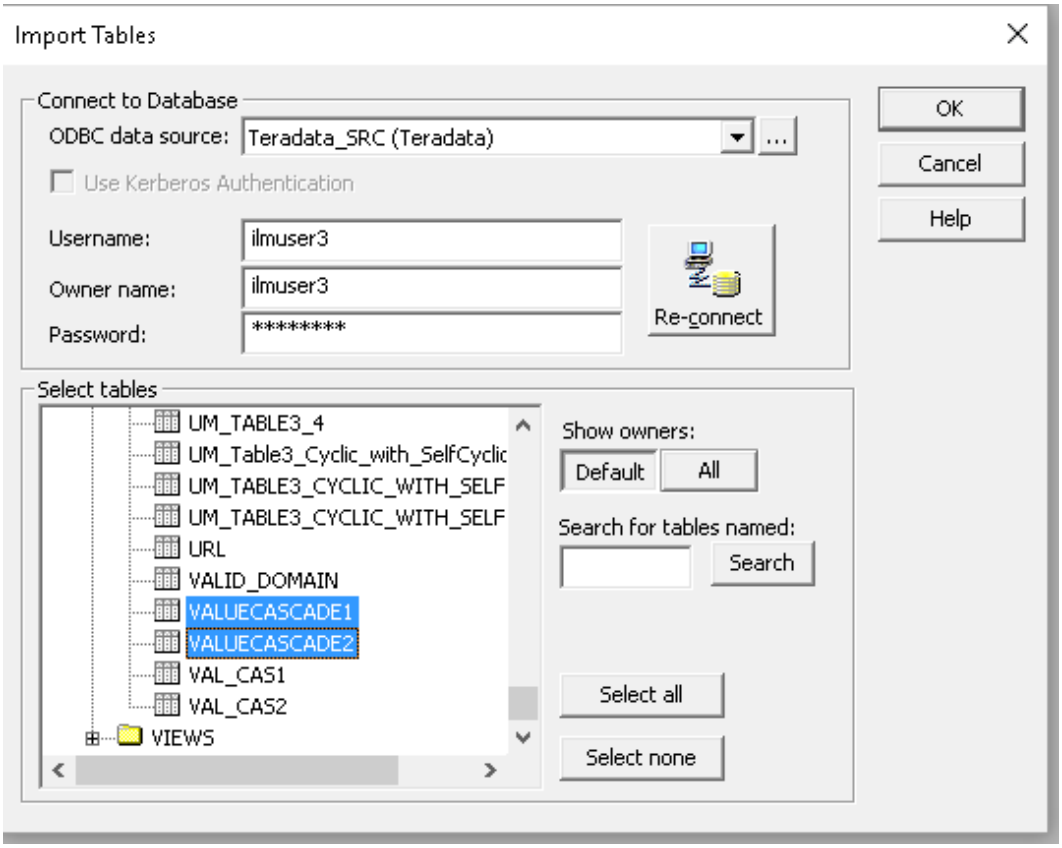
Import the Metadata into PowerCenter

Import source metadata into PowerCenter from Designer.

1. Open the Repository Manager and connect to the required PowerCenter repository.
2. Click **Folder** > **Create**.
3. Enter a name and click **OK** to create the folder.
4. Refresh the repository connection to update the changes.
5. Open the Designer and connect to the required repository.
6. Select the required folder.
7. Click **Sources** > **Import from Database**.

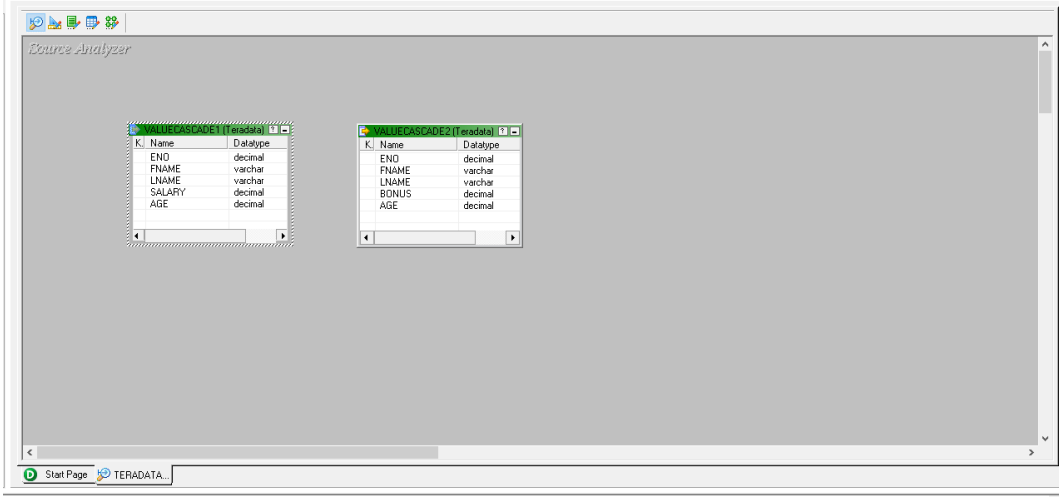
   The **Import Tables** window appears.

The following image shows the **Import Tables** window:



8. Select the Teradata ODBC data source that you created on the PowerCenter Client machine.
9. Enter the user name, owner name, and password and click **Connect**.
10. Choose to show all owners.
11. Select the tables that you want to import and click **OK**.

    The tables are imported into the PowerCenter folder and open in the **Source Analyzer** window. The following image shows the **Source Analyzer** window in Designer:



12. Save the source metadata that you imported.

Create a Project and Import the Metadata into Test Data Manager

Create a project and import the metadata into the project to perform TDM operations on source data.

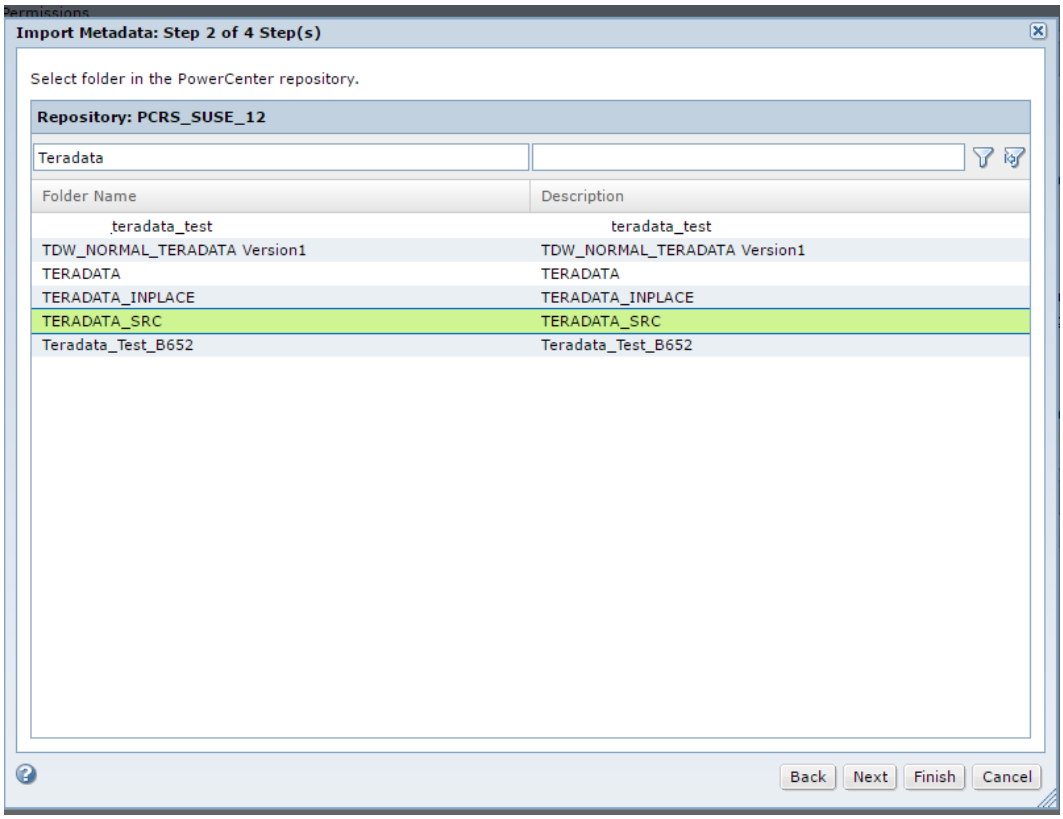
1. Log in to Test Data Manager.
2. Click **Projects** to access the projects.
3. Click **Actions** > **New**.

   The **Create Project** dialog box opens.
4. Enter a name and optional description and click **OK** to create the project.
5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.
6. Choose to import metadata from the PowerCenter repository and click **Next**.

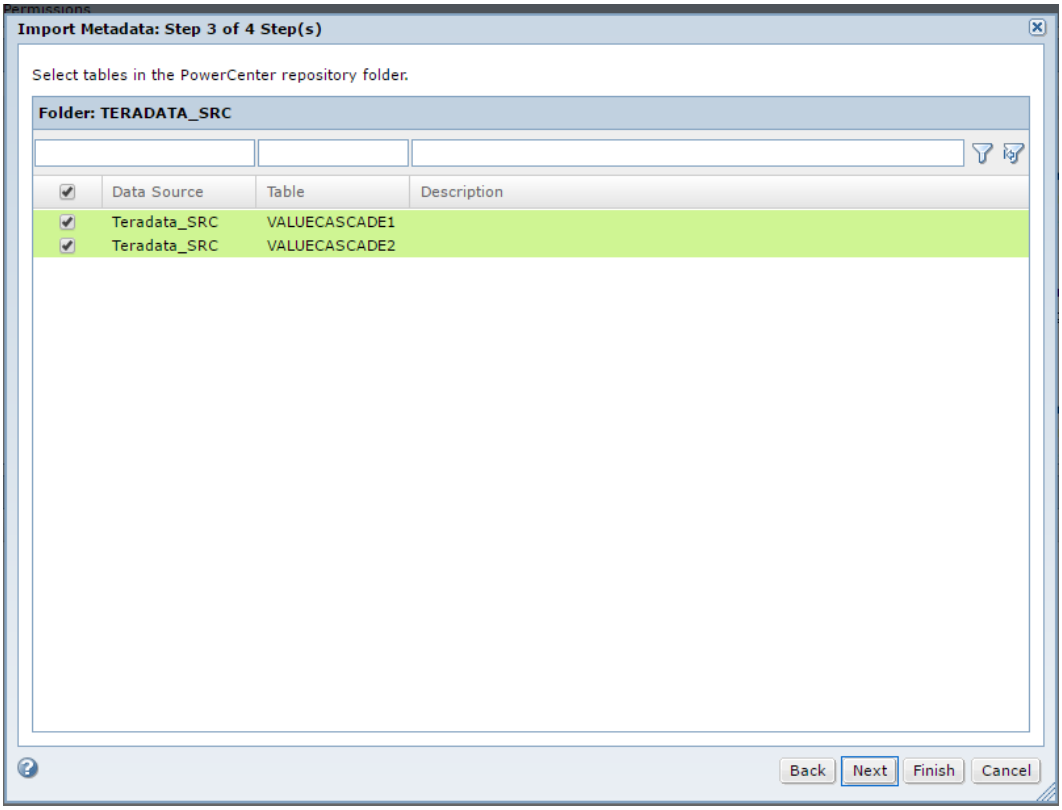
   You cannot directly import metadata from a Teradata connection in Test Data Manager.
7. Select the folder into which you imported the source metadata in Designer and click **Next**.

   The following image shows the second step in the **Import Metadata** wizard:



8. Filter and select the required tables and click **Next**.

The following image shows the third step in the **Import Metadata** wizard:



9. Choose to import the metadata now and click **Finish**.

You can use the source metadata from the Teradata database to perform TDM operations in Test Data Manager.

Create a Teradata Connection in Test Data Manager

Create a Teradata connection in Test Data Manager to use the connection in a plan.

1. In the **Administrator** | **Connections** view, select **New Connection** from the **Actions** menu.

   A window opens to display the new connection properties.

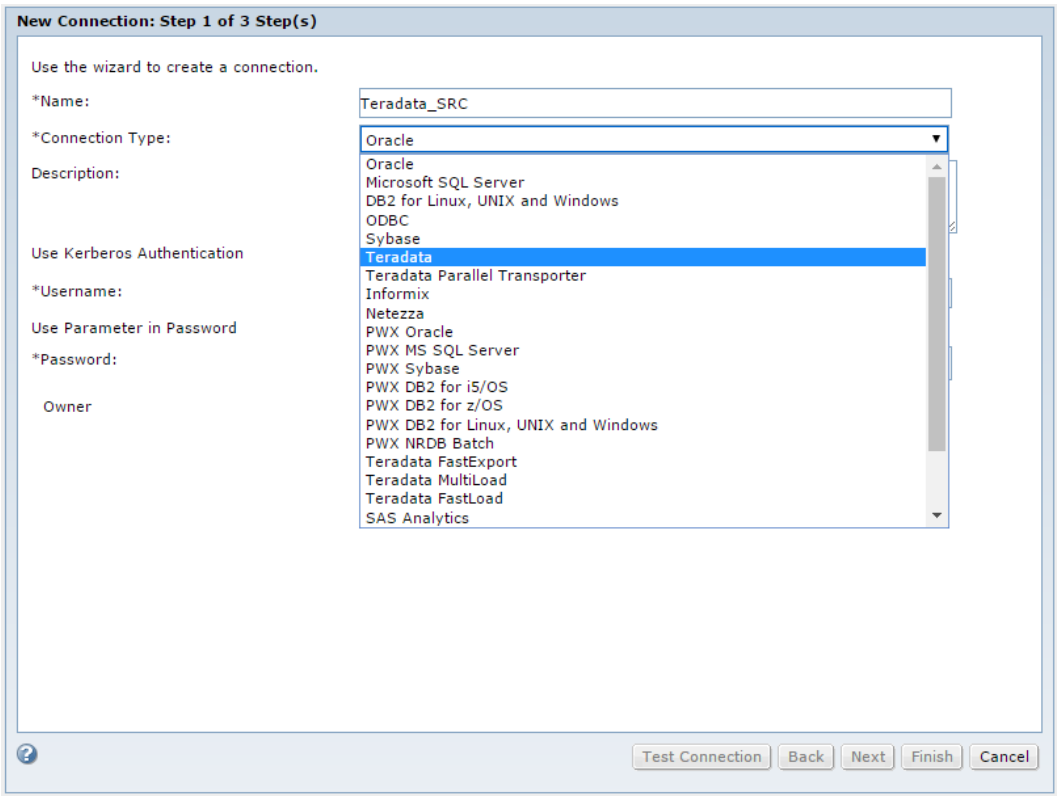
2. Enter a connection name and optional description.

   The connection name must begin with an alphabetic character. If you enter a connection name that begins with a numeric character, a workflow that you use the connection in might fail.

3. Select the Teradata connection that you require and enter the user information. You can create the following Teradata connections:

   - Teradata
   - Teradata Parallel Transporter
   - Teradata FastExport. Use as a source connection only.
   - Teradata FastLoad. Use as a target connection only.
   - Teradata MultiLoad. Use as a target connection only.

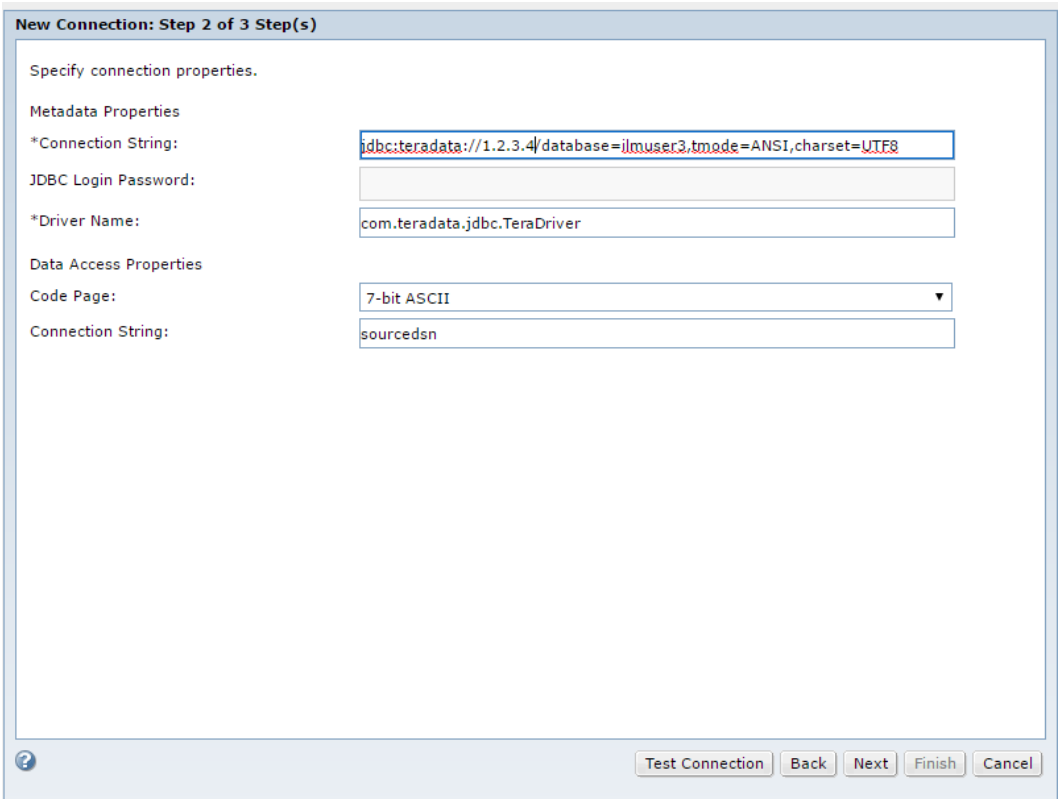
The following image shows the first step of the connection creation wizard:



4. Optional. Click **Change Owner** and select a different user as the connection owner.
5. Click **Next**.
6. Enter the connection properties and click **Next**.

   The connection string must be the name of the connection that you configured in the `odbc.ini` file.

The following image shows the second step of the connection creation wizard:



The properties that appear in step 2 depend on the type of Teradata connection that you select. For information about connection properties, see the *Test Data Management Administrator Guide*.

7. Enter the PowerCenter properties and click **Next**.
8. Optional. Click **Test Connection** to verify that the connection is valid.
9. Click **Finish** to save the connection.

   The connection is visible in the **Administrator | Connections** view.

You can use the connection in a plan.

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